

Space Weather



RESEARCH ARTICLE

10.1029/2021SW002729

Key Points:

- Variability in quiet-time geomagnetically induced currents (GICs) at middle latitudes follows a semidiurnal and annual cycle
- Middle-latitude GIC observations are sensitive to quiet-time magnetic perturbations associated with the Sq current
- GIC quiet-day curves provide a robust baseline for significance analysis of GICs during geomagnetically disturbed times

Correspondence to:

A. C. Kellerman,
akellerman@epss.ucla.edu

Citation:

Kellerman, A. C., Mcgranaghan, R., Bortnik, J., Carter, B. A., Hughes, J., Arritt, R. F., et al. (2022). Geomagnetically induced currents at middle latitudes: 1. Quiet-time variability. *Space Weather*, 20, e2021SW002729. <https://doi.org/10.1029/2021SW002729>

Received 19 JAN 2021

Accepted 16 AUG 2021

Geomagnetically Induced Currents at Middle Latitudes: 1. Quiet-Time Variability

Adam C. Kellerman¹ , Ryan Mcgranaghan² , Jacob Bortnik³ , Brett A. Carter⁴ , Joseph Hughes² , Robert F. Arritt⁵, Karthik Venkataramani² , Charles H. Perry⁵ , Jackson McCormick⁶ , Chigomezyo M. Ngwira² , Morris Cohen⁶ , and Jia Yue^{7,8} 

¹Department of Earth, Planetary, and Space Sciences, University of California Los Angeles, Los Angeles, CA, USA,

²ASTRA, LLC, Louisville, CO, USA, ³Department of Atmospheric and Oceanic Sciences, University of California Los Angeles, Los Angeles, CA, USA, ⁴School of Science, RMIT University, Melbourne, VIC, Australia, ⁵Technical Executive, Electric Power Research Institute, Knoxville, TN, USA, ⁶School of Electrical Engineering, Georgia Institute of Technology, Atlanta, GA, USA, ⁷NASA Goddard Space Flight Center, Greenbelt, MD, USA, ⁸Catholic University of America, Washington, DC, USA

Abstract Geomagnetically induced currents (GICs) at middle latitudes have received increased attention after reported power grid disruptions due to geomagnetic disturbances. However, quantifying the risk to the electric power grid at middle latitudes is difficult without understanding how the GIC sensors respond to geomagnetic activity on a daily basis. Therefore, in this study the question “Do measured GICs have distinguishable and quantifiable long-period and short-period characteristics?” is addressed. The study focuses on the long-term variability of measured GIC, and establishes the extent to which the variability relates to quiet-time geomagnetic activity. GIC quiet-day curves (QDCs) are computed from measured data for each GIC node, covering all four seasons, and then compared with the seasonal variability of thermosphere-ionosphere-electrodynamics general circulation model (TIE-GCM)-simulated neutral wind and height-integrated current density. The results show strong evidence that the middle-latitude nodes routinely respond to the tidal-driven Sq variation, with a local time and seasonal dependence on the direction of the ionospheric currents, which is specific to each node. The strong dependence of GICs on the Sq currents demonstrates that the GIC QDCs may be employed as a robust baseline from which to quantify the significance of GICs during geomagnetically active times and to isolate those variations to study independently. The QDC-based significance score computed in this study provides power utilities with a node-specific measure of the geomagnetic significance of a given GIC observation. Finally, this study shows that the power grid acts as a giant sensor that may detect ionospheric current systems.

Plain Language Summary Changes in the activity of our star, the Sun, can lead to currents flowing through the power grid (known as geomagnetically induced currents, or GICs) at middle latitudes, potentially causing disruptions to our daily lives. In this study, the question “Do measured GICs have patterns over the course of a day, and over different seasons that can be easily identified?” is answered. The study focuses on times when the Sun is quiet, and connects the observed GICs to changes in the upper atmosphere. The results show that indeed, there is a signal in the GICs, which can be connected to atmospheric changes. The study provides a new method which can help power grid operators understand (a) the performance of their current GIC monitors, and (b) whether or not a measured change in current is due to activity driven by the Sun.

1. Introduction

Solar storms result in an eruption of charged particles through flares and coronal mass ejections, which propagate through interplanetary space, and may eventually impact the Earth's magnetosphere. Energy transfer from the solar wind to the magnetosphere drives magnetospheric and ionospheric currents, and consequently, perturbations in the geomagnetic field. These geomagnetic field variations generate a surface geoelectric field via the electromagnetic induction process (Pirjola, 2000; Viljanen & Pirjola, 1994). The generation of these geoelectric fields occurs due to telluric currents flowing through the subsurface structure of the Earth. When the impedance of naturally occurring subsurface material (e.g., rock) is high,

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geomagnetically induced currents (GICs) will preferentially flow through human-made conductors, such as power systems and pipelines, potentially leading to disruption of the power grid. It is therefore necessary to understand how GICs respond to geomagnetic activity in order to mitigate risk (Guillon et al., 2016; Pulkkinen et al., 2005).

Water vapor and ozone absorption of solar radiation in the troposphere and stratosphere results in upward propagating atmospheric tidal waves, leading to global-scale oscillations with harmonic periods of a solar day (Chapman & Lindzen, 1970). These waves drive significant short-term variability in the neutral wind fields within the ionospheric dynamo region from 90 to 150 km (Miyahara & Ooishi, 1997). During quiet times, this variability is manifested in geomagnetic observations and is known as the solar quiet (Sq) variation (Campbell & Matsushita, 1982; Campbell et al., 1993; Graham, 1724a, 1724b; Stewart, 1882; van Sabben, 1964). The neutral wind field is composed of atmospheric tides driven by solar radiation, dragging the electrically conducting fluid through the Earth's magnetic field in the dynamo region, to generate electromotive forces via wind-dynamo theory (Richmond, 1979). The dynamo generates electric currents that manifest as two large-scale vortices centered at middle to low latitudes (below $\sim 60^\circ$)—a counter-clockwise rotation in the northern hemisphere, and a clockwise rotation in the southern hemisphere—and are known as the Sq current.

During strong geomagnetic storms, there have been reports of power grid disturbance at middle latitudes (Koen & Gaunt, 2002; Ngwira et al., 2008; Zois, 2013). Indeed, areas at middle and low latitudes previously considered to be low risk to GICs, have experienced disruptions (Gaunt & Coetzee, 2007), and hence are an important factor when considering the general hazard to the electric power grid (Pulkkinen et al., 2017). Understanding the GIC risk at middle latitudes depends on the nature of the response to geomagnetic activity measured by GIC sensors, including the day-to-day variability which is important for establishing the baseline response.

The question addressed in this study is whether the GICs flowing into transformers at specific locations (nodes) respond to the long-term, quiet-day oscillations, and the extent to which the observed GIC response can be quantified. Quantifying the quiet-day component is important to distinguish the long-term variability from any short-term response, the latter of which may be due to elevated geomagnetic activity. By establishing the long-term baseline response, one can also quantify the significance of any short-term response and establish the relationship to geomagnetic activity.

The structure of the paper is as follows: Section 2 introduces the data sets and quiet-day curve classification criteria; Section 3 investigates the diurnal and annual variability of the GICs and presents a comparison with thermosphere-ionosphere-electrodynamics general circulation model (TIE-GCM) simulations of the neutral wind and height-integrated ionospheric current densities; Sections 5 and 6 discuss and conclude the study.

2. Data Preparation and Quiet-Day Classification

In this study, geomagnetic activity is specified by the SuperMAG auroral-electrojet (SME) (Newell & Gjerloev, 2011) and ring-current (SMR) (Newell & Gjerloev, 2012) indices. SuperMAG is a worldwide collaboration of organizations and national agencies that operate hundreds of magnetometers, and provide access to a unified data set and indices (Gjerloev, 2012). Quiet geomagnetic periods are classified as times where $SME < 100$ nT and $|SMR| < 15$ nT. The GIC data provided for this study are transformer-level observations and part of the SUNBURST network. The Electric Power Research Institute's (EPRI's) SUNBURST network program has proven to be an effective, organized approach for measuring, consolidating, and sharing data related to GIC (EPRI, 2018). The SUNBURST network consists of multiple Hall-effect transducers which are capable of measuring ± 500 Adc (Ampere Direct Current) with $\pm 1.0\%$ accuracy. The present sampling rate is 1-s continuous. A positive sign on the GICs represents current flowing into transformers. The names of individual transformer monitors (nodes) are excluded from this study for proprietary reasons, and instead, nodes are numbered in order of increasing magnetic latitude. The GIC nodes included in this study are located at magnetic latitudes between 40° and 50° .

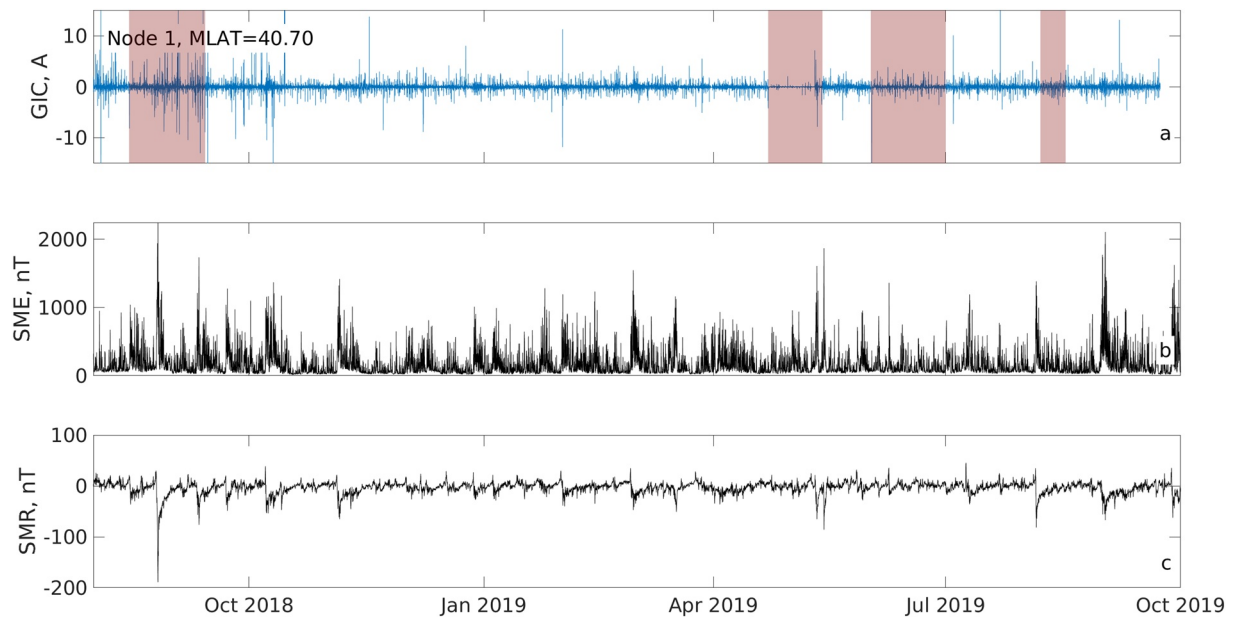


Figure 1. Line plot of measured current flowing into a transformer on the U.S. west coast, (b) SuperMAG auroral-electrojet (SME), and (c) SuperMAG ring-current (SMR) versus time of year. The red shaded region indicates downtime (see text for details). The geomagnetic latitude of the station is given for this node.

Investigation of the GIC data revealed several periods in which the amplitude of current across multiple frequencies in the 1–50 mHz range were lower than other periods. In addition, sudden offsets with associated spikes in the data indicated nongeomagnetic-driven changes in the transformer operation, and immediately preceded or followed the periods of lower amplitude. The period beginning with the sudden offset to a lower amplitude through to the return to “nominal” operation were flagged as “downtime” and removed from further analysis. Figure 1 presents the GIC data set from one US west-coast node, and illustrates a period of downtime by the shaded red region. Figure 2 illustrates the same data processing for several east-coast nodes. The difference in the time period in these two figures are a result of GIC data availability. For this study, data from Node 1 were available from August 2018 to October 2019, while data from Nodes 2 to 7 were available from January 2018 to December 2018.

The cleaned GIC data with downtime removed were averaged over 30-min intervals for each day, and the same 30-min interval from the preceding and superseding 10 days (i.e., 21 days of information), and used to compute percentiles. Several other time intervals and steps were considered, including shorter averaging intervals. The shorter periods resulted in more structure in time, but the variability observed was largely within error (not shown here). The 30-min interval results in a smooth estimation over the course of a day, while the 10-day period provides sufficient points from which to make the quiet-time estimate, especially since elevated geomagnetic activity may occur over the course of several days. Little improvement is gained by including more days, while statistical significance is reduced by lowering the number of days. The same technique was used for each available GIC node to produce a quiet-time curve (QDC) of measured current as a function of magnetic local time, and time of year. An example of the QDC technique applied to Node 7 for a period in July, 2018, is shown in Figure 3. The top panel shows the time of magnetic local time noon by the vertical dotted lines, the measured GIC in gray, and the 10th, 50th, and 90th percentiles by dashed, solid, and dashed lines, respectively. The bottom panel shows the logarithm of the number of valid points (matching the selection criteria) that contributed to the QDC value. The high number of points provides evidence that the derived QDCs are statistically meaningful; the p -values were also found to be near zero, not shown here. The variability in the number of points contributing to the QDC is associated with short-term variability in the measured current, which is not captured within the 21 days, 30 min window used here. A clear diurnal variation is evident in Figure 3a, with a significant prenoon MLT GIC peak, and much of the finer-scale variability bounded by the 10th/90th percentiles. An example of a period of heightened GIC variability is observed on July 28, early in the day, as defined by the GIC exceeding the QDC threshold.

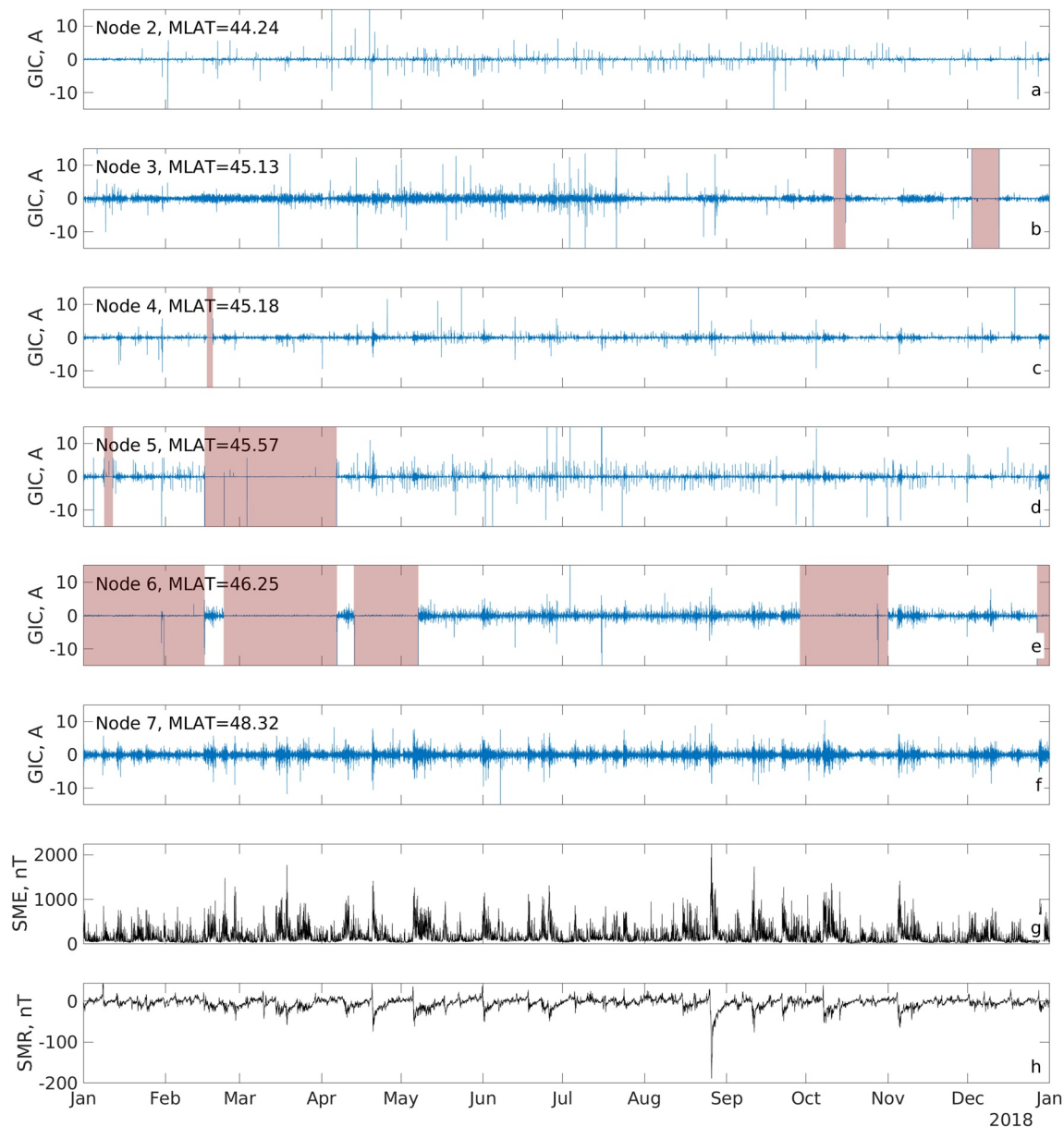


Figure 2. Similar to Figure 1 except for several nodes on the U.S. east coast.

3. Observations and Modeling

Two QDC computations, utilizing the median-based QDC approach, are illustrated in Figures 4 and 5, for the nodes at magnetic latitudes 40.7 and 48.32, respectively. Each figure shows the binned value of (a) solar-zenith angle at the node location, and (b) the GIC QDC, with the binning conducted over the time of year and magnetic local time. The solar-zenith angle (SZA) is representative of the source of energy, the solar heating of the thermosphere, which drives quiet-time variability at middle and low latitudes. The solar heating drives both the diurnal and semidiurnal tides, with the total contribution from semidiurnal tides dominating in the lower ionosphere (Hagan et al., 2001). However, the phase of the semidiurnal tides depends upon altitude, as the waves propagate away from the location of absorption. Hence, we do not look for a one-to-one agreement with SZA and GIC, but rather an observed change with season that implies the solar heating is the primary source of the observed variation.

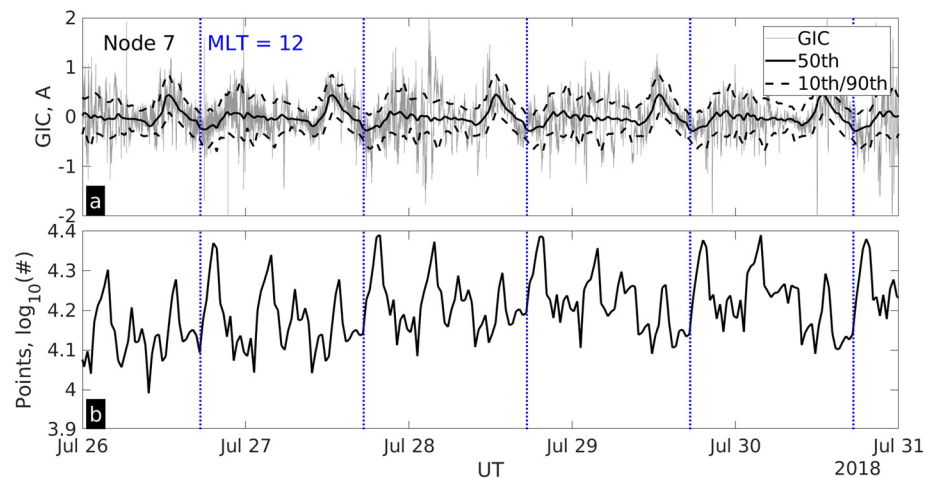


Figure 3. Example of the geomagnetically induced current quiet-day curves (GIC QDC) for Node 7 in July, 2018. The measured GIC is shown by the gray trace. The median, 10th and 90th percentiles are shown in the first panel. The second panel shows the log of the number of points included within each time interval. In both panels, the time of magnetic local time noon is illustrated by the vertical dotted lines.

Figures 4 and 5 display similar QDC distributions, with clear diurnal and seasonal variability in the GIC magnitude. Note that Figure 4 covers the period August 2018 to October 2019, while Figure 5 covers January 2018 to December 2018. In both figures, the peak-positive GIC occurs prenoon. Also evident in both panels is (a) a transition to negative GIC near 12 MLT, and (b) a negative to positive reversal at around 18 MLT. The differences in the time and structure of the GIC QDCs are addressed below. The observed variability in the GIC QDC appears to be semidiurnal, though the peak amplitude appears to be diurnal in nature.

The observed seasonal dependence of the GIC QDCs may match the seasonal dependence of the Sq variation. In order to investigate this in detail, 2-year-long simulations using the TIE-GCM v2.0 model (Qian et al., 2014) were conducted for each of 2018 and 2019. The TIE-GCM is a self-consistent numerical model

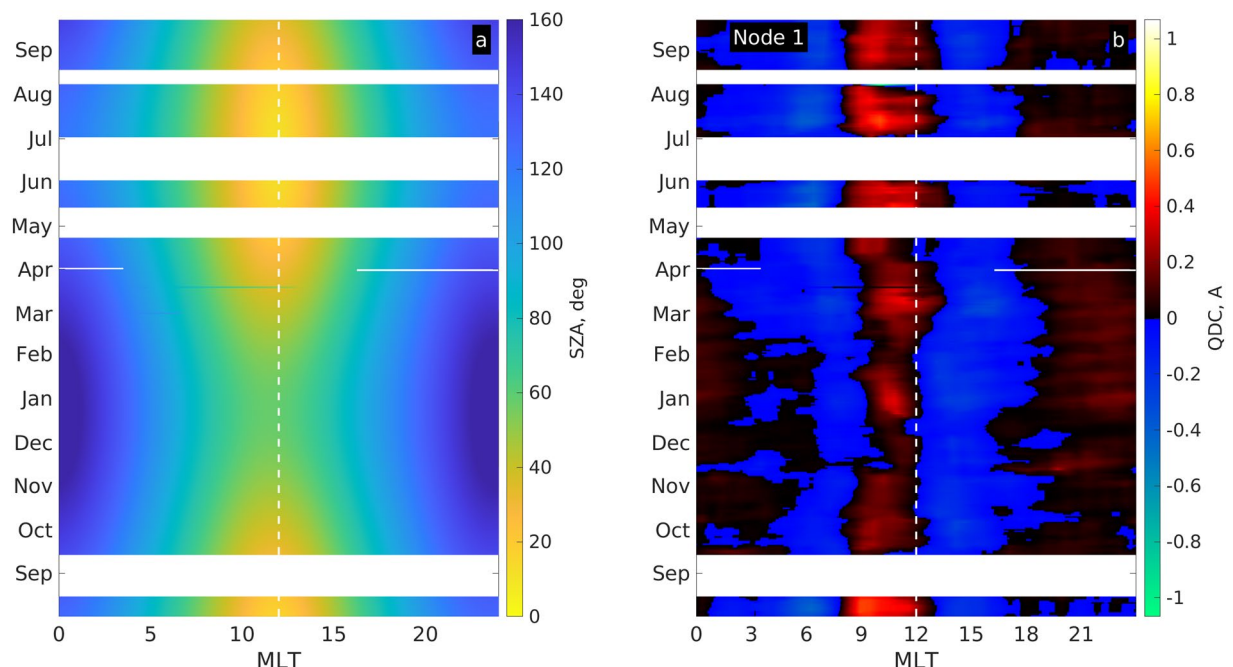


Figure 4. Time versus MLT heat maps for (a) solar-zenith angle (SZA) at Node 1 (b) the Node 1 geomagnetically induced current quiet-day curves (GIC QDC).

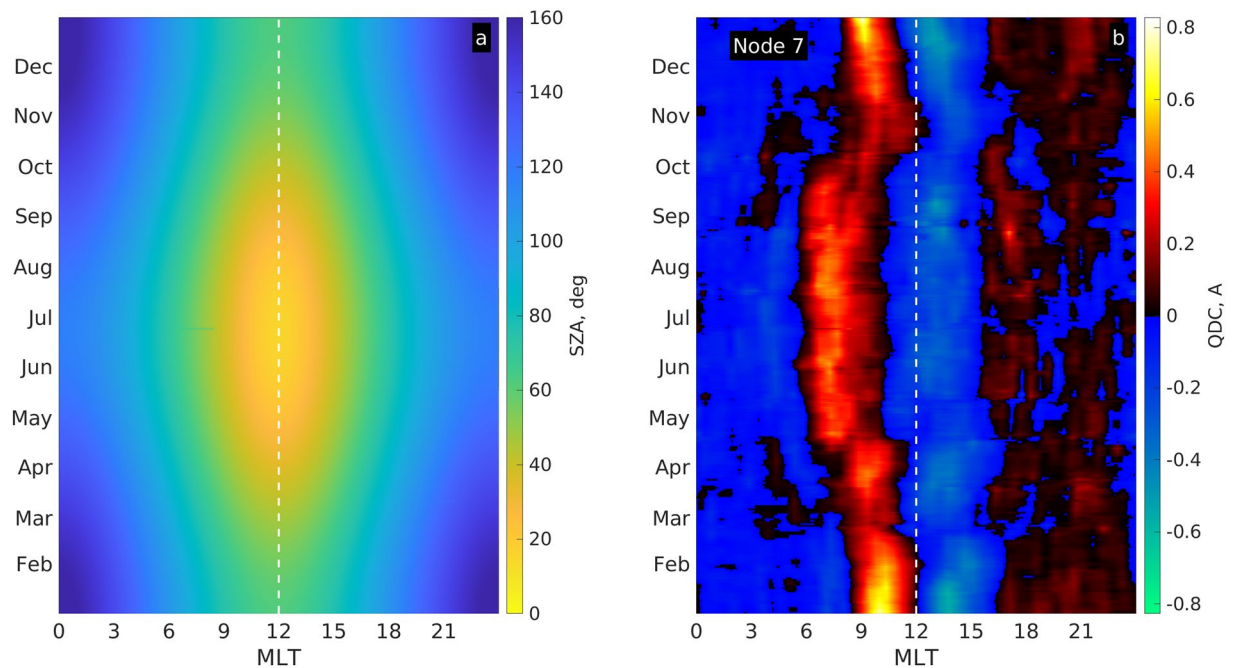


Figure 5. Time versus MLT heat maps for (a) The solar zenith angle (SZ) at Node 7 and (b) The Node 7 geomagnetically induced current quiet-day curves (GIC QDC).

of the thermosphere which includes the dynamics, energetics and chemistry with a steady-state ionospheric electrodynamic in a realistic geomagnetic main field (Richmond et al., 1992; Qian et al., 2014). The model spans from approximately 97 km to 450–600 km, though the model uses log pressure $Z = \ln(p_0/p)$ as the vertical coordinate with the reference pressure p_0 set to $p_0 = 5 \times 10^{-7}$ hPa, and hence varies with solar activity. The ionospheric electrodynamics are calculated in a modified magnetic Apex coordinate system (Richmond, 1995).

In this study, the model runs were conducted at a 5° resolution, using a 60-s time step, a constant $F10.7 = 71$, with $Kp = 2.0$, using climatological GSWM diurnal and semidiurnal migrating tides (Hagan et al., 2001), using the Heelis et al. (1982) high-latitude potential model, with aurora parameterization enabled, using an O-O+ collision factor of 1.5, with electrodynamic enabled forced by the wind dynamo, and gravity and plasma pressure gradient driven current; the height-integrated ionospheric current is determined; calculation of helium was enabled. The model outputs were linearly interpolated to the location of each node for the analysis that follows.

Figure 6 displays the height-averaged (100–120 km geometric height) TIE-GCM-modeled meridional neutral wind and measured GIC QDC at Node 4. One should note that meridional wind strength and direction varies with altitude and so the comparison here is only presented to show that there is a correlation between the neutral wind and the GIC QDC; below, the full height-integrated currents are considered. Figure 6 shows that in the northern hemisphere winter months, the MLT-dependent transitions in the meridional wind correspond well with transitions in the GIC QDC, albeit with some small differences. In the northern hemisphere summer months, however, the early afternoon and dusk transitions occur at similar MLT (the 12–24 MLT sector), however the morning transitions in GIC QDC exhibit more structure, including a peak in the GIC QDC, that is, not associated with a negative meridional wind.

The tidal-driven wind-dynamo drives currents in the ionosphere leading to perturbations at the Earth's surface (Richmond, 1979), which may drive the low amplitude GICs observed in Figure 6. The neutral winds vary in direction and magnitude as height increases into the dynamo region, and hence the average wind field may not capture combined effects leading to GIC in the power grid. Further, the differences in the observed GIC at each location are a function of the complexity and orientation of the interconnected power grid, and subsurface conductivity structure. Therefore, the contribution to the GIC may be better analyzed

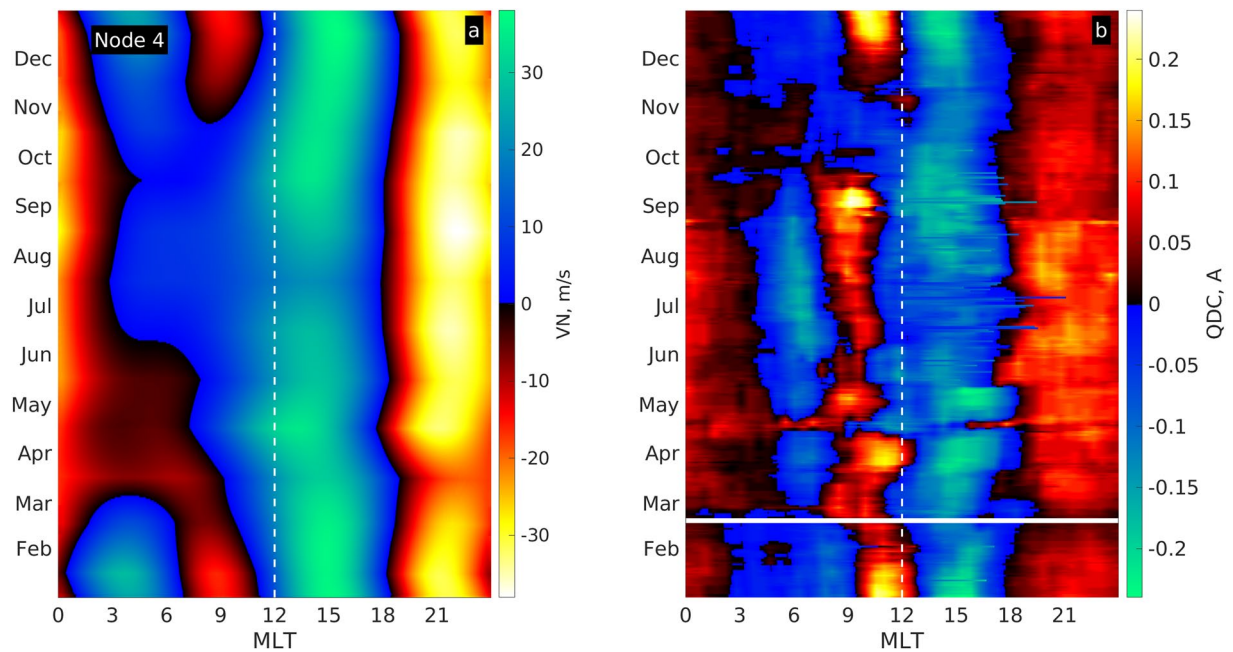


Figure 6. Binned value plot of (a) Thermosphere-Ionosphere- Electrodynamics General Circulation Model (TIE-GCM)-modeled meridional neutral wind velocity (VN) and (b) geomagnetically induced current quiet-day curves (GIC QDC) at Node 4. Each quantity is plotted versus time of year and magnetic local time. Neutral winds from 100 to 120 km geometric height are considered.

by considering the influence of components of the height-integrated horizontal currents. The TIE-GCM model allows to compute height-integrated current densities $K_{q\phi}$ and $K_{q\lambda}$ (Richmond, 1995), representing the east and north directions, respectively. The height-integrated current above each node is assumed to be representative of the wider-area ionospheric current contribution as the model resolution is $5 \times 5^\circ$, and variation in the current as a function of MLT and season within the region surrounding each node is insignificant during the quiet times considered in this study. Ultimately, the height-integrated current above each node is thus representative of the wider ionospheric contribution to the geoelectric field at each node location. One would have to sample model output 1,000's of km away to see a significant change in the observed MLT and seasonal variation of the integrated currents.

A comparison between the TIE-GCM-modeled height-integrated horizontal current density and the measured GIC QDC is presented in Figure 7 for five nodes. We seek a method to account for potential differences in the aforementioned grid configuration and subsurface conductivity structure that is specific to each node. Therefore, influence and orientation factors are introduced, which determine the importance of either the north-south or east-west current in driving the observed GIC QDC. We introduce a new scalar factor X , which we call an influence factor defined as $X = a(K_{q\phi}) + b(K_{q\lambda})$, with $|a| + |b| = 1$, and where a and b specify the relative importance of the east-west and north-south components, respectively, as well as the direction (positive east and north). The introduction of these factors allows one to find the importance of either east-west or north-south components of the current, and hence determine the orientation most likely to drive positive GICs at each node. The a and b are indicated for each node in Figure 7, and found numerically. The numerical solution minimizes the mean-absolute error between the normalized GIC QDC and the influence factor, X .

Figure 7a shows X at Node 1, with numerically derived factors $a = -0.13$ and $b = -0.87$, which indicate that a south-west directed current leads to a positive GIC at this node. Further, we expect a 13% influence from the west current and 87% from the south current, which suggests that meridional currents dominate the GIC response at this node. The corresponding GIC QDC is shown in Figure 7b, and the absolute error of the normalized GIC and X is shown in Figure 7c. The general shape of the positive X resembles the GIC from 9 to 24 MLT, with a narrow region of enhanced X in winter, and a wider region in the northern hemisphere

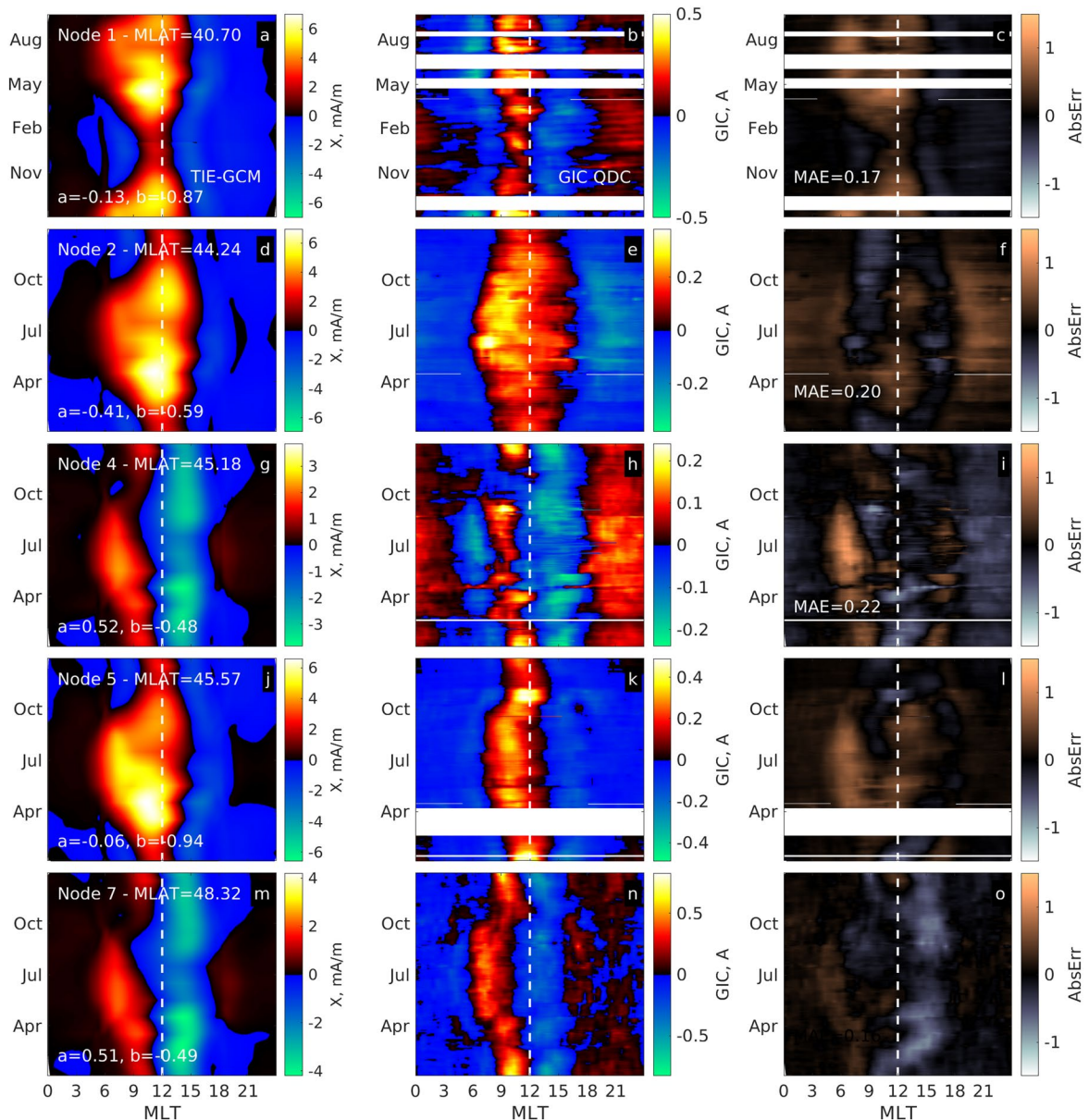


Figure 7. Binned value plot of the influence factor (X), geomagnetically induced current quiet-day curves (GIC QDC), and the absolute error between the normalized values of X and the GIC QDC. The mean absolute error is shown for each node. See text for details.

summer. In the summer-morning sector (0–9 MLT), the agreement is not well aligned in MLT, with a noticeably wider region of positive X . The mean absolute error between the best fit X and GIC QDC is 0.17.

Figure 7d displays X at Node 2. Similar to Node 1, a south-west current leads to a positive GIC at this node. In this case, the region of positive X generally matches that of the GIC QDC in Figure 7e, though there are some discrepancies between the value of X post dawn and in the early afternoon, with X again showing an earlier transition in the northern hemisphere summer.

Figure 7g displays X at Node 4. In this case, a south-east directed current leads to a positive GIC, with near equal influence from south and east currents ($a = 0.52$ and $b = -0.48$). The transitions in X match those from the GIC in Figure 7h during the northern hemisphere winter months, however in the summer months, the GIC exhibits more structure post dawn as seen by the strong positive absolute error in Figure 7i

near 6 MLT. The discrepancy is similar to what was seen in Figures 6 and 7c. The agreement between the X and GIC QDC transitions is good from the prenoon to dusk sector, with absolute error values less than 1.

Figure 7j displays X at Node 5. A south-west current leads to a positive GIC at this node, with a dominant influence from the meridional component. Similar to Figure 7a, the transitions in X match those from the GIC QDC in Figure 7k during the northern hemisphere winter months, however in the summer months the extent of X across MLT is larger and, in Figure 7l, a clear positive error near 6 MLT is observed.

Finally, Figure 7m displays X at Node 7, with a south-east current leading to a positive GIC. Similar to Figure 7g, nearly equal ($a = 0.51$) east and ($b = -0.49$) south current influence factors are found to best align X and GIC. The agreement between X and the GIC QDC in Figure 7n is very good across all months, as borne out by the low absolute error values in Figure 7o, although the X does turn positive at a slightly earlier MLT than the GIC QDC in northern hemisphere summer, and the negative X appears to be too strong for the observed GIC QDC response.

In summary, Figure 7 illustrates a striking similarity between the influence factor, X , and the GIC. A clear relationship is evident between the diurnal variation and seasonal variability of X and the GIC QDCs. The direction and magnitude of the zonal (east) and meridional (north) components are shown to be a critical factor determining the quiet-day response of these nodes. This directional dependence may also provide insight into how each GIC node may respond during active times. At nodes where north-south currents dominate the GIC ($|a| \ll |b|$), the MAE indicated that, in northern hemisphere summer, a northward current may not be fully captured by the TIE-GCM model near 6 MLT; a region of negative GIC was routinely observed for each of these nodes in Figure 7. In the next section, the QDC-GIC values are employed in an example event to demonstrate their value during geomagnetically active times.

4. GIC During a Geomagnetically Active Period

In this section, the quiet-time component of the GIC response is used to compute a baselined GIC response, which contains only the active-time component. The baselined GIC data are then used to infer the true impact of geomagnetic activity on the GIC response for a specific period of time. The baselined GIC value is thus $GIC_{BL} = GIC_{obs} - QDC_{50}$, where GIC_{obs} are the observed GIC values, and QDC_{50} is the 50th percentile of the QDC distribution. The significance is computed by way of a interpercentile range (IPR) z score, defined in terms of the tenth or 90th percentile as follows:

$$z_{IPR} = \begin{cases} GIC_{BL} / \delta_{90}, & \text{if } GIC_{obs} \geq 0 \\ GIC_{BL} / \delta_{10}, & \text{otherwise} \end{cases}$$

where $\delta_{90} = QDC_{90} - QDC_{50}$ and $\delta_{10} = QDC_{50} - QDC_{10}$. The subscripts 10 and 90 refer to the time-dependent 10th and 90th percentiles of the QDC distribution. The z_{IPR} hence provides a robust measure of significance for either negative or positive GIC observations, which reliably and repeatably isolates the GMD-related GIC.

Figure 8 presents a moderate storm period in August 2018. The data were obtained from the OMNI database, and represent quantities propagated to the Earth's bow shock. Three events are highlighted in Figure 8, in green, blue, and red. The first event in green identifies a period of calm prior to the storm. The second event in blue highlights a sharp increase in dynamic pressure, which is shown in Figure 8a. The third event in red highlights the sharp increase in SME, which is shown in Figure 8c, and a period of elevated solar wind velocity, as shown in Figure 8b. Figure 8d presents the observed GIC from Node 6 for this period, with the QDC 10, 50, and 90 percentiles over plotted. The baselined GIC values are shown in Figure 8e, and the significance score is shown in Figure 8f.

In Figure 8d, the observed GIC exhibits peaks of $\sim 2A$ in events 1 and 3, and a negligible response in event 2. Figure 8d illustrates that the quiet-time component contributes $\sim 1A$ to GIC in event 1, and $\sim 0A$ in events 2 and 3. The baselined GIC GIC_{BL} in Figure 8e represents the active component of the GIC response, and illustrates that the active component contributes $\sim 1A$ to event 1, approximately $0.5A$ to event 2, and nearly the entire $2A$ to event 3. One can conclude that the GIC response observed in event 3 was due to enhanced geomagnetic activity. The significance of the GIC response, as it pertains to geomagnetic active time, is

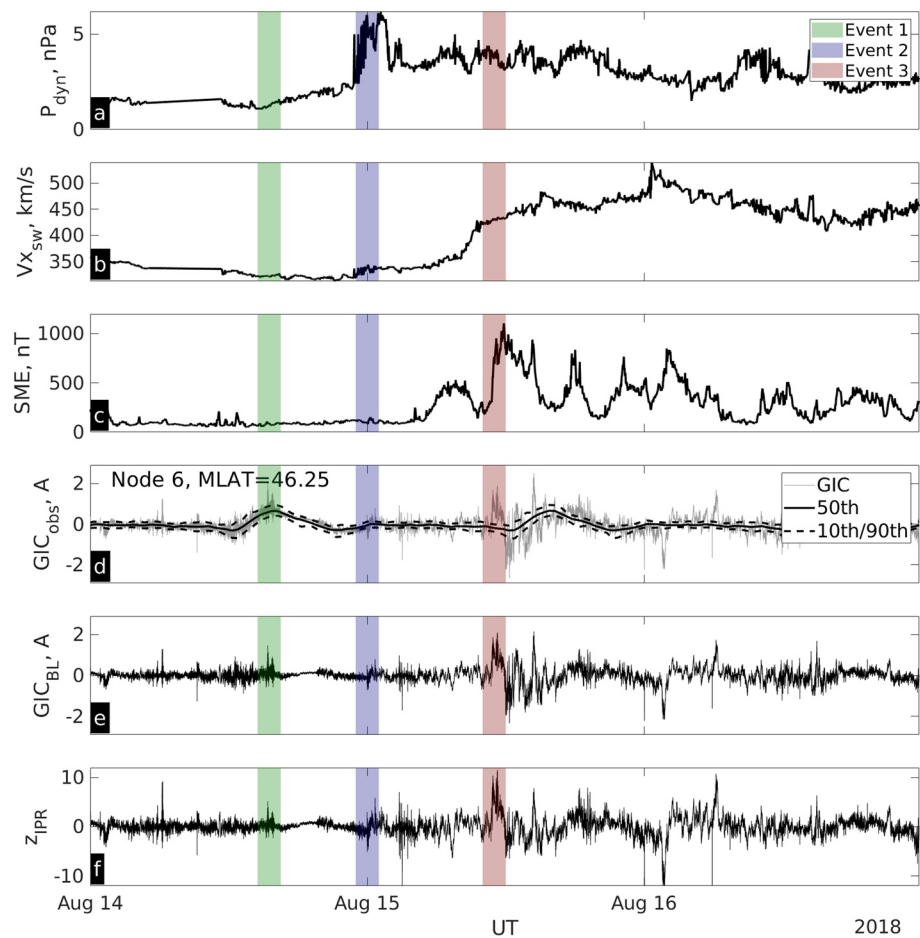


Figure 8. Line plots of a moderate storm period for Node 6, displaying (a) dynamic pressure, (b) solar wind velocity, (c) SuperMAG auroral-electrojet (SME), (d) the geomagnetically induced current quiet-day curve (GIC), (e) quiet-day curve-baselined GIC, and (f) interpercentile range z score, versus UT. Three events of interest are highlighted in green, blue, and red.

illustrated in Figure 8f. The z score confirms that the variability in event one is much less significant (~ 5) than event 2 (~ 10), with regards to elevated geomagnetic activity.

One may have noticed other periods where the significance score was elevated, usually very briefly. These spikes correspond to short periods of elevated GIC, which are understood to largely correspond to nongeomagnetic impacts. However, at the time of writing this manuscript the exact source of these spikes is unknown, and therefore the data are still included in our analysis. Our focus on event 3 in the above analysis was due to the sustained elevated significance that was observed near August 15, 12 UT for a period exceeding an hour. Other periods of short, elevated significance were observed in only one or two nodes, suggesting that they were due to a more localized source, and not wide-spread geomagnetic activity.

The significance scores for nodes 2–7 are presented together in Figure 9 for the same period of time. For each node, the significance is higher during event 3 than event 1, revealing that enhanced geomagnetic activity is responsible for a significant response across all nodes. Interestingly, several nodes also show a significant response during event 2, simultaneous with the increase in dynamic pressure. Although only a single period has been shown here, these results demonstrate that the QDCs are a powerful data set from which the impact of geomagnetic activity on the electric power grid can be quantified.

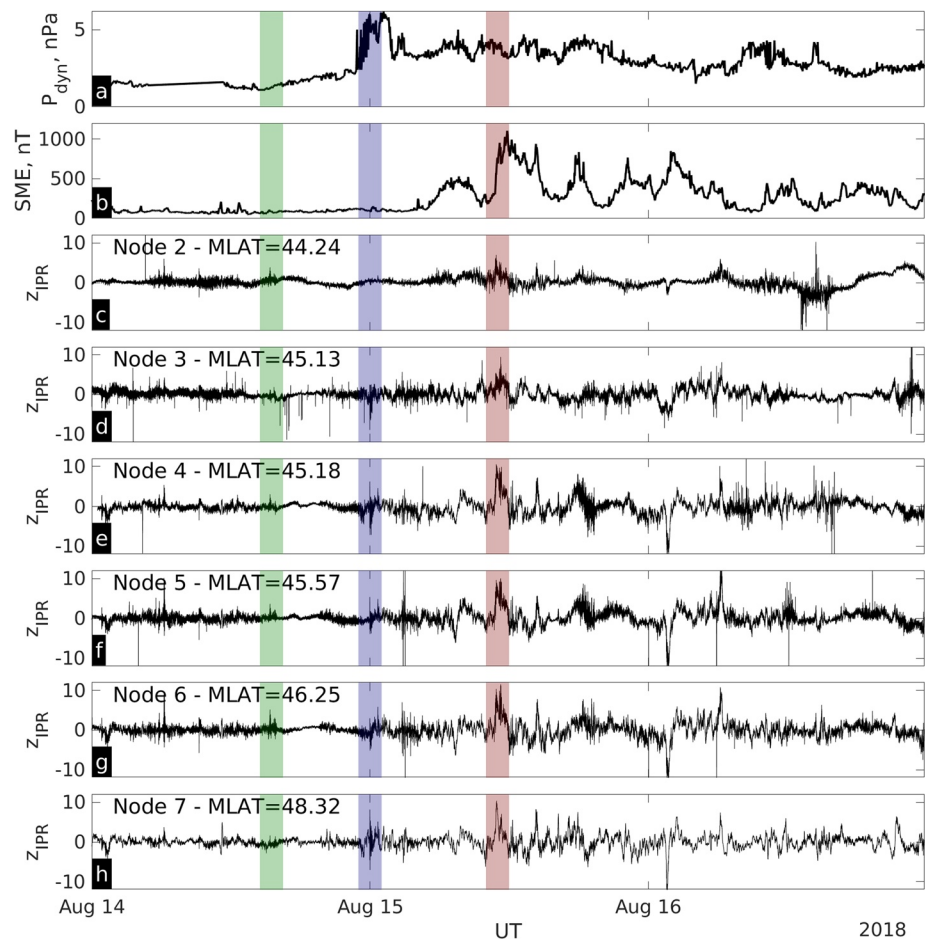


Figure 9. Line plots of (a) dynamic pressure, (b) SuperMAG auroral-electrojet, and (c–h) significance score for Nodes 2–7. See text for details.

5. Discussion

The TIE-GCM simulations in Figures 6 and 7 illustrate the good agreement between the tidal-driven Sq current and the observed quiet-day perturbations in the GIC detected at several transformer locations at middle latitudes. The identified dependence on the direction of the height-integrated horizontal currents provides a node-specific estimate of the current direction that would lead to a positive GIC at each node location. The magnitude of the response exhibits a clear MLT dependence, which is important for accurately specifying the effect of geomagnetic storms on GIC currents, as shown in Figures 8 and 9.

A discrepancy between X and the quiet-day component of the GIC is identified in northern hemisphere summer, near dawn for several nodes, where a positive X corresponds to a negative quiet-day GIC; a transition to positive quiet-day GIC occurs later each morning. Given the orientation flags used ($b < 0$), this suggests that a period of stronger northward current may not be fully represented in the TIE-GCM model during northern hemisphere summer, and in the postdawn sector. Similarly, recent work has shown that a thin-sheet model approach was unable to fully reproduce the quiet-time variability of GIC (Bailey et al., 2017). However, the overall good agreement between X and the GICs here supports the idea that the migrating tidal-driven perturbations in ionospheric currents are responsible for the quiet-time variability in the GICs.

The dependence of each node on a particular orientation of the height-integrated current density, reflected in X , is consistent with the notion that information on the grid configuration is potentially as important as the GIC observations themselves. For instance, Nodes 1 and 5 show similar seasonal variability and

orientation flags, though Node one is located on the U.S. west coast at 40.7° MLAT, while Node five is located on the east coast at 45.57° MLAT. Whereas Nodes 4 and 5 show very different seasonal variability, though both nodes are located on the east coast within 1° MLAT of each other. Indeed, past analysis of the GIC response at middle-latitude in New Zealand revealed different transformer-level GIC responses within the same substation (Mac Manus et al., 2017). The dependence of the GIC QDC on X shown in Figure 7, comprising the variable importance factors, a and b , confirms the strong influence of the grid configuration on the observed GIC response.

Since global geomagnetic disturbances comprise the quiet (Sq) and disturbance (Dst) components of the geomagnetic field, the good agreement between QDC GICs and the X variation suggests that the QDCs provide a suitable (Sq) baseline from which to define the significance of the observed GIC during active times, and hence a robust method to quantify the effects of geomagnetic disturbance (GMD) on the electric power grid. This methodology was applied to investigate the GIC response during one time period, as shown in Figures 8 and 9. The analysis demonstrated that the largest, active-component GIC measured during this period could be attributed to perturbations at $40\text{--}80^\circ$ MLAT, demonstrating the capability to estimate the significance of geomagnetic activity on any observed GIC. The techniques applied to derive a QDC for the GIC may be applicable to any node, however, the specifics of the analysis considered in this study are most applicable to latitudes equatorward of the auroral region, and poleward of the equatorial fountain.

This section will conclude by noting several aspects of the current work which are useful for power utilities: (a) By defining the expected GIC behavior during quiet times, QDC's provide a robust baseline which operators may use to ensure their GIC monitors are operating nominally, (b) Identification of the dominant ionospheric current direction leading to positive GIC at each node can help understand the risk to that node during an active event (theoretical or real, and provided the grid configuration is the same), (c) GIC modeling may be tested during quiet times, as we have a very good understanding of the geomagnetic drivers and the expected GIC response - this may help to improve GIC modeling during active times. (d) The significance score computed in this study provides power utilities with a measure of the geomagnetic significance of a given GIC perturbation, which could be applied in real time. Finally, we expect the QDCs derived here will be applicable to times other than the 2018–2019 period studied here, as the inputs to the TIE-GCM model were based on statistical averages, not specific to any particular year.

6. Conclusions

In this study, GIC measurements from several middle-latitude nodes and SuperMAG geomagnetic indices were used to develop quiet-day curves for geomagnetically induced currents. The potential driver of the observed variability in each QDC was investigated by employing the TIE-GCM model, leading to the following conclusions:

1. Quiet-time GIC observations exhibit quantifiable variability that depends on magnetic local time and season.
2. The quiet-time GIC matches well with daily and seasonal changes in height-integrated horizontal currents above each node location, and hence are attributed to the Sq current system. The direction of the currents is an important factor in determining the influence of the ionospheric currents on any given GIC node. These results show that the power grid may be used as a giant sensor for ionospheric currents at middle latitudes.
3. Given that GIC nodes respond to the quiet-time Sq variation, the GIC QDCs may be used as a robust baseline to define the significance of GIC measurements during geomagnetically disturbed times.

Data Availability Statement

The data provided for this study are part of the SUNBURST network, and were obtained by working with EPRI on this study. For the current study, the authors have made the GIC data and computed QDC levels from Nodes 1–7 available on Zenodo (<https://zenodo.org/record/4444068>).

Acknowledgments

The authors gratefully acknowledge the SuperMAG collaborators (<https://super-mag.jhuapl.edu/info/?page=acknowledgement>) for the indices employed in this study available at <https://super-mag.jhuapl.edu/indices>. The authors acknowledge use of NASA/GSFC's Space Physics Data Facility's OMNIWeb service, and OMNI data available at <https://omniweb.gsfc.nasa.gov>. We acknowledge the developers of the TIE-GCM model, available at <https://www.hao.ucar.edu/modeling/tgcm/tie.php>. Simulation analysis was assisted by the Community Coordinated Modeling Center at Goddard Space Flight Center (<http://ccmc.gsfc.nasa.gov>). The authors acknowledge the developers of the IRBEM library, which was utilized to obtain magnetic coordinates in the study. A. C. Kellerman and J. Bortnik were supported by NSF grant 1937152. R. M. Mcgranaghan was supported by NSF grants 1937152 and 1940208. B. A. Carter was supported by the Australian Research Council's Linkage Project scheme (LP160100561). M. Cohen and J. McCormick were supported by NSF grant 1937152.

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